

100 Practical Programming Practice Questions

Curriculum Focus: C++ Loops and Text Input

Part 1: Introducing for Loops

1. Write a program that uses a for loop to print the numbers from 1 to 100.
2. Write a program to print the numbers from 100 down to 1 using a for loop.
3. Write a program that calculates and prints the sum of all integers from 1 to 50.
4. Write a program that asks the user for an integer N, then prints all even numbers from 2 to N.
5. Write a program to print all odd numbers between 1 and 100.
6. Write a program that prints the multiples of 5, up to 100.
7. Write a program to calculate the factorial of a user-provided positive integer using a for loop.
8. Write a program that asks the user for a word and a number N, then prints the word N times.
9. Write a program that prints the alphabet from 'a' to 'z' using a for loop.
10. Write a program that prints the alphabet in reverse order from 'Z' to 'A'.
11. Write a program to compute and display the squares of the numbers from 1 to 15.
12. Write a program to compute and display the cubes of the numbers from 1 to 10.
13. Write a program that uses a for loop to print the first 20 terms of the Fibonacci sequence.
14. Write a program that takes an integer input and prints its multiplication table up to 10.
15. Write a program to calculate X^Y (X raised to the power of Y) using a for loop, where X and Y are provided by the user.
16. Write a program that prints every third number starting from 1 up to 60.

17. Write a program to calculate the alternating sum: $1 - 2 + 3 - 4 + \dots + N$.

18. Write a program that counts and prints how many numbers between 1 and 200 are divisible by 7.

19. Write a program that asks for 5 integers from the user in a loop and prints their average.

20. Write a program to print the ASCII values of the characters 'A' through 'Z'.

Part 2: The while Loop

21. Write a program that prints the numbers from 1 to 20 using a while loop.

22. Write a program that continuously asks the user for integers and adds them up, stopping when the user enters 0.

23. Write a program that asks the user for a positive number. If they enter a negative number or zero, use a while loop to keep asking until they provide a valid input.

24. Write a program that finds and prints the first power of 2 that is greater than 10,000.

25. Write a program that counts the number of digits in a user-provided integer.

26. Write a program that reverses the digits of an integer (e.g., turns 456 into 654) using a while loop.

27. Write a program that calculates the sum of the digits of a given integer.

28. Write a program to find the Greatest Common Divisor (GCD) of two numbers using a while loop.

29. Write a program that prompts the user for a password and uses a while loop to keep asking until they type 'secret'.

30. Write a program that simulates a countdown timer from 10 to 1, then prints 'Liftoff!'.

31. Write a program that takes a number and repeatedly divides it by 2 (integer division) until it becomes 0, printing each step.

32. Write a program that generates the Collatz sequence for a given starting integer N until it reaches 1.

33. Write a program to calculate the Least Common Multiple (LCM) of two numbers using a while loop.

34. Write a program that reads floating-point numbers from the user until a negative number is entered, then prints the maximum value entered.

35. Write a program that checks if a given integer is a palindrome (reads the same forwards and backwards).

Part 3: The do while Loop

36. Write a menu-driven program that displays 3 options. Use a do while loop to keep showing the menu until the user selects the 'Exit' option.

37. Write a program that simulates rolling a 6-sided die. Use a do while loop to roll again if the user types 'y'.

38. Write a program that strictly requires the user to input a number between 1 and 10. Use a do while loop to enforce this.

39. Write a basic calculator that adds two numbers, displays the result, and asks 'Do you want to perform another addition (y/n)?'.

40. Write a program to print the numbers 1 to 5 using a do while loop.

41. Write a 'Guess the Number' game where the target is hardcoded to 42. Loop until the user guesses correctly.

42. Write an ATM PIN entry simulation. Allow the user a maximum of 3 attempts to enter the correct PIN (1234).

43. Write a program that calculates the running total of inputted numbers, but the loop is guaranteed to execute at least once even if the first input is 0.

44. Write a program that continuously prompts for two integers, prints their sum, and stops only when both integers are 0.

45. Write a program using a do while loop that prints characters from 'a' to 'z' separated by commas.

46. Write a program that reads student grades. Stop when the user enters -1, then calculate the average grade.

47. Write a program that asks for a string input and repeats it back, stopping when the user types 'quit'.

48. Write a program to print powers of 3, starting at 1, stopping when the value exceeds 1000.

49. Write a program that keeps asking the user to flip a coin (by entering 'h' or 't') until they get three 'h's in a row.

50. Write a Rock-Paper-Scissors game that plays one round and asks if the user wants to play again.

Part 4: The Range-Based for Loop (C++11)

51. Write a program that initializes an array of integers {10, 20, 30, 40, 50} and prints all elements using a range-based for loop.

52. Write a program that calculates the sum of all elements in an array of double values using a range-based for loop.

53. Write a program to find the maximum value in a pre-defined integer array using a range-based for loop.

54. Write a program to find the minimum value in an array using a range-based for loop.

55. Write a program that counts how many even numbers are in an array using a range-based for loop.

56. Write a program that iterates over a string literal and prints each character on a new line.

57. Write a program that calculates the average of an array of floats using a range-based for loop.

58. Write a program that searches an array for a specific value provided by the user and prints whether it was found.

59. Write a program that prints all array elements separated by commas.

60. Write a program that uses a reference variable in a range-based for loop to multiply every element in an integer array by 2.

61. Write a program that converts all characters in a string to uppercase using a reference in a range-based for loop.

62. Write a program that counts the number of vowels in a given string using a range-based for loop.

63. Write a program that finds and prints the first negative number in an array, then breaks out of the loop.

64. Write a program that concatenates all elements of an array of strings into a single string.

65. Write a program that copies all elements from one integer array to another using a range-based for loop.

Part 5: Loops and Text Input

66. Write a program using `cin.get()` in a while loop that reads characters until the EOF is reached and counts the total number of characters.

67. Write a program that reads text input character by character and counts the number of space characters.

68. Write a program that reads text input and counts the number of newline characters.

69. Write a program that reads characters and replaces all spaces with underscores before printing them.

70. Write a program that reads words until the user enters the word 'STOP'.

71. Write a program that reads characters from input and converts lowercase letters to uppercase before echoing them.

72. Write a program that counts the number of sentences in a text input by counting the occurrences of punctuation.

73. Write a program that reads integers from standard input until EOF is reached, and then prints their sum.

74. Write a program that reads text and echoes it to the screen, but omits all vowels.

75. Write a program that reads characters and checks if parentheses are balanced.

76. Write a program that reads a block of text and counts how many times the letter 'e' appears.

77. Write a program that reads characters, ignores any numeric digits, and prints the remaining text.

78. Write a program that reads text and capitalizes the first letter of every word.

79. Write a program that reads a single word and uses a loop to print it backwards.

80. Write a program that reads multiple words from input and identifies the longest word based on character count.

Part 6: Nested Loops and Two-Dimensional Arrays

81. Write a program that uses nested loops to print a 5x5 square of asterisks (*).

82. Write a program that uses nested loops to print a right-angled triangle of asterisks with a height of 5.

83. Write a program that uses nested loops to print an inverted right-angled triangle of asterisks.

84. Write a program that uses nested loops to print a 10x10 multiplication table.

85. Write a program that initializes a 3x3 two-dimensional array with zeros and uses nested loops to print it.

86. Write a program that calculates the sum of all elements in a hardcoded 3x4 2D array.

87. Write a program to find and print the maximum element in a 4x4 array.

88. Write a program that calculates and prints the sum of each individual row in a 3x3 array.

89. Write a program that calculates and prints the sum of each individual column in a 3x3 array.

90. Write a program that calculates the sum of the main diagonal elements in a 4x4 array.

91. Write a program that calculates the sum of the secondary diagonal elements in a 4x4 array.

92. Write a program that transposes a 3x3 array (swaps rows and columns) and prints the new array.

93. Write a program that checks if a 3x3 array is symmetric.

94. Write a program that multiplies every element of a 2x2 array by a scalar value provided by the user.

95. Write a program that adds two 2x2 arrays and stores the result in a third 2x2 array.

96. Write a program that multiplies two 2x2 arrays and stores the result in a third array.

97. Write a program that uses nested loops to print a checkerboard pattern using 'X' and 'O'.

98. Write a program that searches for a specific user-provided value in a 5x5 array and prints its coordinates.

99. Write a program that uses nested loops to print a centered pyramid pattern of asterisks.

100. Write a program that reads a 2x3 two-dimensional array and flattens it into a 1D array of size 6.